

Row4 = Row4
- 2 × Row2

$$\begin{bmatrix} 2 & -4 & -2 & 3 \\ 0 & 3 & 1 & -1 \\ 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & -5 \\ 0 & -9 & -3 & 13 \end{bmatrix}$$

Row5 = Row5
+ 3 × Row2

$$\begin{bmatrix} 2 & -4 & -2 & 3 \\ 0 & 3 & 1 & -1 \\ 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & -5 \\ 0 & 0 & 0 & 10 \end{bmatrix}$$

~~$$L = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$~~

[Note: It will turn out that A has only three pivot columns, so the method of Example 2 will produce only the first three columns of L. The remaining two columns of L come from

$\begin{bmatrix} 5 \\ 5 \end{bmatrix}$

$$\begin{bmatrix} 2 \\ 6 \\ 2 \\ 4 \\ -6 \end{bmatrix} \div 2$$

$$\begin{bmatrix} 3 \\ -3 \\ 6 \\ 9 \end{bmatrix} \div 3$$

$$\begin{bmatrix} 5 \\ -5 \\ 10 \end{bmatrix} \div 5$$

$$L = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ -3 & 3 & 2 & 0 & 1 \end{bmatrix} \quad \square$$